

Program : Diploma in Instrumentation Engineering	
Course Code : 4081	Course Title: Process Variables Measurement
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To impart students with the fundamental concepts of pressure measurement and its applications in industries.
- To explain the construction, working principle and applications of different types of flow meters.
- To give an insight about different techniques and applications of level measurement.
- To illustrate the fundamental concepts and applications of different types of temperature measuring techniques.

Course Prerequisites:

Topic	Course code	Course name	Semester
Concept of transducers		Sensors and transducers	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Compare the construction, working principle and applications of different types of pressure measuring instruments.	15	Applying
CO2	Explain the construction, working principle and applications of different types of flow meters.	15	Understanding
CO3	Explain the different techniques for level measurement.	12	Understanding

CO4	Apply different types of transducers for temperature measurement	16	Applying
	Series Test	2	

CO- PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2						
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Compare the construction, working principle and applications of different types of pressure measuring instruments.		
M1.01	Convert between different pressure standards.	2	Applying
M1.02	Describe the construction and working principle of different types of manometers	3	Understanding
M1.03	Explain the construction and working principle of elastic pressure gauges	3	Understanding
M1.04	Compare and apply different transducers for various pressure measurements.	3	Applying
M1.05	Explain different methods used for vacuum measurement	2	Understanding
M1.06	Calibrate various pressure measuring instruments.	2	Understanding

Contents:

Definition of pressure - different types of pressure - units of pressure – simple conversion problems-Methods of pressure measurement- manometers- U tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems-Elastic pressure transducers-C-type and spiral bourdon gauges - Bellows – Diaphragms-Electrical pressure transducers- Strain gauges- Capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- Differential pressure transmitter -Vacuum gauges- McLeod gauge, Pirani gauge - Ionization gauge -Testing and Calibration of pressure measuring instruments - dead weight tester- manometers –pressure transmitters. I/ P and P/I converters using flapper nozzle system.

CO2	Explain the construction, working principle and applications of different types of flow meters.		
M2.01	Recall the fundamentals of flow measurement.	4	Understanding
M2.02	Compare the construction and working principle of different types of flow measurement techniques.	8	Understanding
M2.03	Calibrate various flow measuring instruments.	3	Understanding
	Series Test - I	1	
Contents: Principles of Flow measurements- Laminar and turbulent flow – Reynold's number - Continuity equation - Bernoulli's equation – simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method.			
CO3	Explain the different techniques for level measurement.		
M3.01	Explain the construction and working principle of different types of level measurement techniques.	8	Understanding
M3.02	Compare and apply different transducers for various level measurements.	4	Understanding
Contents: Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type – hydro static pressure type- . Air purge type level indicator – electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge – microwave Level switches – Differential Pressure type level transmitter – simple problems.			
CO4	Apply different types of transducers for temperature measurement		
M4.01	Convert different temperature scales.	3	Applying
M4.02	Explain the construction and working principle of different types of temperature measurement techniques	8	Understanding
M4.03	Compare and apply different transducers for various temperature measurement	5	Applying
	Series Test - I	1	

Contents:

Definition of temperature - different temperature scales and conversions – simple problems.

Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer-Definition of PTC and NTC – construction and working of Resistance Temperature Detector – different types (Ni, Cu, & Pt) - characteristics of RTD – simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect – laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- E,J,T,K,S,R - properties of different types– simple problems-thermopile - Compare the characteristics of Thermocouple, RTD and thermistor.

Text / Reference:

T/R	Book Title/Author
R1	A.K. Sawhney, A course in Mechanical measurements and Instrumentation, Dhanpat Rai
R2	S.K Singh , Industrial Instrumentation and Control , TMH
R3	R.K Jain , Mechanical and Industrial measurements , Khanna Publishers
R4	Krishnaswami , Industrial instrumentation , New Age Publications

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=4knAR4HzRJs
2	www.temperatures.com › Vendors › Electronics › A To D
3	https://www.youtube.com/watch?v=As5kzxkyT24
4	https://www.youtube.com/watch?v=K2CH1cUkMgs
5	https://www.youtube.com/watch?v=JHCS7ldf4aU
6	nptel.ac.in/courses/101106040/chapter 5.pdf
7	https://www.youtube.com/watch?v=-MQU0xgh6bA
8	https://instrumentationtools.com/dp-transmitters-level-measurement
9	https://www.emerson.com/.../measurement-instrumentation/level